

CIRCUITX SMART SOLVER

Final Semester Project Report

Student	Juman Chowhan
University	Sukkur IBA University
Department	Computer Systems Engineering
Semester	III
Subject	Circuit Analysis (CA)
Instructor	Engr. Asif Ali

1. Introduction

CircuitX Smart Solver is a Java-based educational software project for basic circuit analysis. It is designed to help students calculate voltage, current, resistance, power, and related quantities using common circuit-analysis formulas. The current version is a console-based prototype and will be expanded during the final semester project.

2. Problem Statement

Students frequently perform electrical calculations manually while studying circuit analysis. Repeated calculations can take time and small arithmetic mistakes can occur. CircuitX aims to provide a simple and structured software tool that performs selected calculations and displays the formula and result.

3. Objectives

- Develop a simple Java-based circuit-analysis solver.
- Calculate electrical quantities using standard formulas.
- Apply programming concepts to an engineering problem.
- Provide a foundation for a larger circuit-analysis application.
- Add more analysis techniques during future development.

4. Current Scope and Features

The current prototype includes Ohm's Law, Power Calculator, Series Resistance, Parallel Resistance, Voltage Divider Rule (VDR), and Current Divider Rule (CDR). The software uses a menu-driven console interface.

5. Technologies and Programming Concepts

The project is currently implemented in Java. It uses Scanner for user input, methods for separate calculation modules, if/else conditions, loops, arrays, switch-case, and basic mathematical operations.

6. Main Formulas

Ohm's Law: $V = I \times R$, $I = V / R$, $R = V / I$.

Power: $P = V \times I$, $P = I^2 \times R$, $P = V^2 / R$.

Series Resistance: $R_{eq} = R_1 + R_2 + \dots + R_n$.

Parallel Resistance: $1/R_{eq} = 1/R_1 + 1/R_2 + \dots + 1/R_n$.

VDR: $V_x = V_s \times R_x / R_{eq}$.

CDR: $I_x = I_t \times R_{eq} / R_x$.

7. Working Methodology

The program starts with a main menu. The user selects a calculation method, enters the required electrical values, and CircuitX applies the relevant formula. The result is then displayed. After completing a calculation, the program returns to the main menu until the user selects Exit.

8. Current Implementation Status

The current code is a basic prototype. It focuses on implementing the core calculations rather than providing a graphical interface or solving arbitrary circuit diagrams. This version is being uploaded to

GitHub to document the project's development progress.

9. Limitations

The current version is console-based. It does not yet solve arbitrary circuit diagrams, automatically analyze complex networks, or provide a graphical circuit editor. Input validation is also basic. Some advanced circuit-analysis techniques are not implemented yet.

10. Future Work

Planned improvements include KCL, KVL, mesh analysis, nodal analysis, Thevenin and Norton theorems, additional network theorems, a Java GUI, circuit-diagram support, stronger input validation, and support for more complex circuit configurations.

11. Conclusion

CircuitX Smart Solver provides a starting point for a larger educational circuit-analysis application. The current prototype demonstrates how Java programming can be applied to common electrical calculations. The project will be expanded step by step with additional analysis methods and software features.

12. Repository Contents

The GitHub repository contains the current Java source code and README documentation. Screenshots and additional project documentation can be added as the project develops.